Abdelmalek Essadi University

Algebra IV — Final Exam

Year: 2022

Exercise 1:

Let $\alpha \in \mathbb{R}$. Consider the matrix $A_\alpha \in M_3(\mathbb{R})$ defined by:

$$A_\alpha = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ -\alpha & 1 & 1 \end{pmatrix}$$

1. Determine the **characteristic polynomial** of the matrix A_α.
2. For which values of α is the matrix A_α **diagonalizable** over $\mathbb{R}$?
3. Consider the equation in $M_3(\mathbb{R})$:

$$M^2 = A_\alpha \quad (*)$$

For which values of α does this equation $(*)$ possess **4 diagonalizable solutions** over $\mathbb{R}$?

Answer Area

Exercise 2:

Consider the matrix $A \in M_3(\mathbb{R})$, defined by:

$$A = \begin{pmatrix} -1 & -2 & 0 \\ 2 & 3 & 0 \\ 2 & 2 & 1 \end{pmatrix}$$

1. Determine a **Jordan reduction relation** for the matrix A.
2. Deduce a **Jordan reduction relation** for the matrix $-A^{-1}$.
3. Determine the polynomials $\mathcal{Q} \in \mathbb{R}[X]$ such that the matrix $\mathcal{Q}(-A^{-1})$ is **diagonalizable**.

Answer Area

Exercise 3:

Let $\alpha \in \mathbb{R}$. Consider the matrix $A_\alpha \in M_3(\mathbb{R})$, defined by:

$$A_\alpha = \begin{pmatrix} 2\alpha+2 & \alpha+1 & 1 \\ -4\alpha-2 & -2\alpha-1 & -2 \\ 2\alpha-1 & \alpha-1 & 0 \end{pmatrix}$$

1. (a) Determine $\mathrm{Rg}(A_\alpha - I_3)$.
   (b) Deduce $N_\alpha(X)$.
   (c) Without performing any calculation, what is the value of $\dim E_{-1}(A_\alpha)$?
2. For which values of $\alpha \in \mathbb{R}$ is the matrix A_α **diagonalizable**?
3. Write the **minimal polynomial** of A_α.
4. We choose $\alpha = 1$ and set $A = A_1$.
   (a) Determine a **Jordan reduction relation** for A.
   (b) Determine the polynomials $P \in \mathbb{R}[X]$ such that A and $P(A)$ are **similar**.
5. (a) Determine a **diagonalization relation** for A_0.
   (b) Without performing any calculation, what is the value of A_0^k, where $k \in \mathbb{N}$?
   (c) Determine the matrices $M \in M_3(\mathbb{R})$ that satisfy $M^2 = A_0 + I_3$ and $\mathrm{Tr}(M) > 0$.

Answer Area